Hong Kiat, Tan

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Education

University of California, Los Angeles

- **Ph.D.** Candidate, Mathematics. Aug 2021 – Dec 2026
- Advisor: Andrea L. Bertozzi. (GPA: 3.98/4.00)
- **Master of Arts** in Mathematics. Aug 2021 – Jun 2022

National University of Singapore

- **Bachelor of Science (Honours)** in Applied Mathematics with Highest Distinction. Aug 2017 – May 2021
- Second Major in Physics and Minor in Statistics. (GPA: 4.97/5.00)
- *Ho Family Prize* – Top graduating student in Applied Mathematics, with 28 A+'s in Math/Physics/Statistics courses.

Industrial/Work Experiences

Applied Scientist Intern, Amazon Search — Jun 2026 – Present

- Developing statistically principled RL methods and ranking theory for post-training LLM-based search rankers.
- Ran ∼60,000 H100-hours of multi-node training and inference to date.

Quantitative Research Intern, WorldQuant Intraday Team — Sep 2025 – Dec 2025

Data Scientist Intern, Amazon Search Data Science and Economics — Jun 2025 – Sep 2025

- Built a framework orchestrating multi-agent LLMs with ℓ^0-changepoint detection for 78 search metrics in 17 locales to generate interpretable economic insights. (ICLR TSALM Workshop 2026.)
- Built agentic production pipeline on AWS Bedrock AgentCore using Strands/LangChain with ECS/Docker orchestration.

Academic Experiences

Mentor & Research Fellow, Supervised Program for Alignment Research — Feb 2026 – Present

- Introduced Activation Flow, a training-free, differential-geometric method that manufactures steering activations from answer labels alone, recovering 60% of the sandbagging accuracy gap from one label and 99% from 40. (Submitted.)
- Proposed a causal model of sandbagging in the residual stream to explain why steering works, and validated it across 11 LLMs (1.5–32B) and 4 lock types (prompting, fine-tuning, RL, circuit-breaking). (2 x AAAI submissions.)
- Developed Circuit Oracle, a training-free multi-agent system that answers safety-relevant queries over LLM circuits through composable skills and custom tools. (ICML CompLearn & Mechanistic Interpretability Workshops, 2026.)

Graduate Research Assistant, UCLA — 2022 – Present

- Proved generic structural stability of Riemann solutions for systems of hyperbolic conservation laws by generalizing Thom's parametric transversality theorem to a foliated setting and introducing sequential transversality, a new differential-topology concept for causally chained submanifolds. (SIAM J. Math. Anal., 2026, sequel in preparation.)
- Developed a modified vector autoregressive framework using double machine learning and causal inference for confounding-adjusted lag detection in time series, utilizing EconML, TabPFN, and PyTorch in Python.
- Architected an object-oriented Python framework coupling game-theoretic models and PDE solvers on directed graphs, optimizing evacuation routing on a non-smooth loss. (Discrete and Continuous Dynamical Systems, 2027.)

Graduate Teaching Assistant, UCLA — 2021 – 2025

- Developed 786 pages of instructional materials across 10 quarters for advanced mathematics courses (Algorithms, Probability, Graduate PDEs, Mathematical Finance, and Analysis), with an average teaching evaluation of 8.6/9.0.

Undergraduate Research Assistant, NUS — 2020 – 2021

- Developed a numerical scheme with hypothesis testing and regression in R for quantum field theory. (Phys. Rev. D, 2022.)
- Co-authored a 148-page paper on a fundamental conjecture in mathematical general relativity. (arXiv:2402.16250, 2024.)

Undergraduate Research Assistant, UNC – Chapel Hill — 2019

- Designed Bayesian hierarchical models for astrophysical data with MCMC samplers in R. (Astrophys. J., 2020.)

Technical Skills & Professional Activities

- *Programming Languages*: Python, C++ (Intermediate), PostgreSQL, R, LaTeX.
- *Agentic Tooling*: Skills, MCPs, Anthropic/OpenAI/Claude Code SDKs, Building Agentic Harnesses.
- *Cloud & DevOps*: AWS (AgentCore, Lambda, ECS/ECR, Bedrock), GCP, Modal, RunPod, Docker, Linux (SLURM), Git.
- *Libraries/Frameworks*: PyTorch, vLLM, verl, SGLang, Hugging Face, wandb, TabPFN, EconML, sklearn.
- *Coursework*: High-dim Statistics, Differential Geometry, Optimization, NLP, Causality, Functional Analysis, PDEs.
- *Service*: Reviewer for AISTATS 2026, UAI 2026, NeurIPS 2026, ICML Workshop 2026.